

DOCUMENTATION
TRAWELL TECHNICAL DOCUMENTATION

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1. INTRODUCTION

Trawell is a web-based platform designed to assist users in creating, organizing, and managing personalized travel itineraries. The system integrates modern web technologies to provide a secure, responsive, and intuitive experience. The project leverages a full-stack architecture built using [Next.js](#), Node.js, Firebase authentication, MongoDB, Mongoose, TailwindCSS, and Jest for testing.

This document provides a formal overview of the system's functionality, architecture, installation process and technical components.

2. PROJECT OBJECTIVES

The main objectives of this project include:

- Developing a reliable and user-friendly interface for travel itinerary management.
- Implementing secure authentication and session handling through Firebase.
- Designing a scalable backend with MongoDB and Mongoose.
- Ensuring maintainability through modular architecture and custom hooks.
- Integrating automated testing to ensure correctness and stability.
- Deploying the application on Vercel with continuous integration.

3. TECHNOLOGY STACK

Technology	Purpose
Next.js 14	Framework for server-side rendering, routing, and performance optimization
React	Component-based UI development
Node.js	Execution environment for APIs and backend logic
Firebase authentication	User management and secure authentication
MongoDB	NoSQL database for persistent storage
Mongoose	ODM tool for scheme and model management
TailwindCSS	Utility-based styling and responsive design
Jest + React Testing Library	Unit and integration testing
Vercel	Production deployment and hosting

4. SYSTEM ARCHITECTURE

Trawell follows a modular full-stack architecture, consisting of:

A) Frontend layer

Developed using [Next.js](#) and React, providing:

- Server components and client components following [Next.js](#) App Router standards.
- Dynamic pages for itinerary creation, editing, and visualization.
- Auth-protected routes using Firebase Authentication.
- Reusable UI components styled with TailwindCSS

B) Backend layer

The backend is composed of:

- [Next.js](#) API Routes for server-side logic.
- A connection layer to MongoDB, set up through Mongoose.
- CRUD operations for itinerary management
- Validation and error handling mechanism

C) Authentication layer

Powered by Firebase, handling:

- Account creation.
- Sign-in and sign-out.
- Session persistence.
- Secure communication with protected API endpoints.

D) Database layer

- MongoDB stores itineraries using structured Mongoose models.
- Ensures scalability and flexibility for the future schema modifications.

5. FEATURES

- Secure login and authentication.
- Creation, editing, and deletion of travel itineraries.
- Persistent cloud storage for user data.
- Responsive and modern user interface.
- Modular and maintainable components and hooks.
- Integration with external APIs (future expansion).
- Testing setup using Jest.
- Production deployment via Vercel.

6. INSTALLATION AND SETUP



Clone the repository

Shell

```
git clone https://github.com/juan-camilo-rosero/trawell.git  
cd trawell
```

Install dependencies

Shell

```
npm install
```

Environment variable

Create a .env.local file with:

Textproto

Firebase Configuration

```
NEXT_PUBLIC_FIREBASE_API_KEY=  
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=  
NEXT_PUBLIC_FIREBASE_PROJECT_ID=  
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=  
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=  
NEXT_PUBLIC_FIREBASE_APP_ID=
```

MongoDB

```
MONGODB_URI=
```

API Base URL

```
NEXT_PUBLIC_API_BASE_URL=
```

Run development server

Shell

```
npm run dev
```

Application runs at:

- <http://localhost:3000>



Build for production

```
Shell
npm run build
npm star
```

7. TESTING

Trawell includes automated tests using Jest and React Testing Library.

Run all test

```
Shell
npm test
```

Watch mode

```
Shell
npm run test:watch
```

Testing includes:

- Unit and integration testing.
- Workflow and interaction testing

8. PROJECT STRUCTURE

```
Shell
/src
├── app/
│   ├── api/                # Backend API routes
│   ├── (routes)/           # Public and private UI pages
│   └── layout.tsx          # Root layout
├── components/             # Reusable React components
├── hooks/                  # Custom React hooks
└── lib/                    # Firebase, DB utilities
```



```
|— models/                # Mongoose models
|— __tests__/             # Jest tests
```

9. SYSTEM WORKFLOW

- a. The user opens Trawell and log in using Firebase Authentication.
- b. Firebase provides a secure user token (UID).
- c. The frontend sends authenticated requests to [Next.js](#) API routes.
- d. API routes validate the session and access MongoDB via Mongoose.
- e. CRUD operations are performed on itineraries.
- f. Responses update frontend state (often via custom hooks)
- g. UI renders itinerary data dynamically and updates in real-time.

10. ACKNOWLEDGMENTS

This project was developed as part of an academic and practical initiative exploring full-stack web development using modern technologies.

We acknowledge the contributor and the open-source libraries that made this project possible.